

PSP DB cross-region replication with Goldengate

Scope

- Build cross region DR for PSP database in AWS US-EAST using GoldenGate replication.
- Refresh AWS non-prod RDS with production data on demand

Setup

PSP application make use of two database - Monolith and IBOB.

Below fine points describes how PSP RDS setup in US-EAST differs from one in US-WEST :

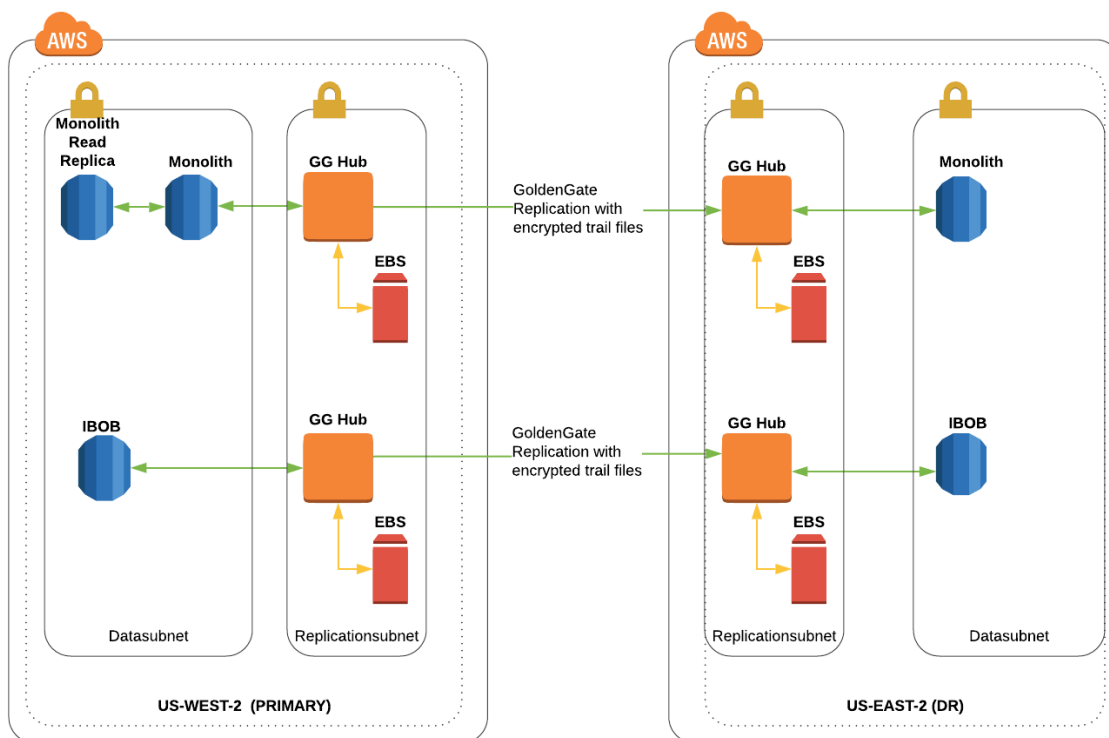
- No Multi-AZ
- 7 days backup in DR region compare to 35 days in primary region.

In terms of subnets, it would be same as primary - RDS in DataSubnet and GoldenGate HUB in ReplicationSubnet

Data security

- PII and sensitive data are encrypted right on application layer using IDPS encryption
- One time data copy to DR would be done by doing database snapshot backup copy across region (as we use data encryption at rest for RDS, snapshot backup would already be encrypted)
- Further data replication would be done using GoldenGate replication which would make use of encrypted trail file across the region
- Connectivity to RDS in US-EAST would be using Oracle Native Network Encryption (NNE) to ensure secure data transmission between client and server

Setup diagram



Review items

- Data security risk for one time snapshot backup copy across region
- Data security risk for ongoing replication across region

- Data security risk of transferring/restoring RDS snapshots from production to non-prod account
- Assess data security risk of running performance/benchmark test on production data in non-prod environment